Hackathon Submission (Level-1 Solution)
Use Case Title: **Culbi**
Team Name: YourAnswer
Institution: Universitas Tanjungpura

Data:
https://theaseanpost.com/article/awareness-lacking-within-asean-0

https://knowledge.csc.gov.sg/special-edition-dec-2020/singapore-asean-and-cultural-competence/

https://www.britishcouncil.org/sites/default/files/regional_perception_poll_asean_creative_economy.pdf

https://ayoacademy.org/importance-of-cultural-understanding-in-asean/#:~:text=Introduction,making%20(Tan%2C%202023).

1. **Problem Statement**

   ASEAN is home to a rich mosaic of cultures, yet regional fragmentation persists due to a profound lack of deep cultural exchange. This is clearly reflected in the data: **50% of stakeholders** identify the lack of regional coordination and cooperation as the primary challenge hindering a cohesive ecosystem. While **60% of the public** perceives the region's creative economy as growing, a significant "cultural illiteracy" gap remains. Only **21% of respondents** are truly familiar with the broader creative and cultural landscape, while the majority, **47%**, hold only a surface-level understanding of it.

   This gap is made worse by rigid educational systems. Currently, **53% of creative and cultural programs** still rely on traditional formal education, which often struggles to translate regional nuances into real, lived experiences. As a result, young people are left with shallow perceptions of their neighbors, and the development of essential soft skills — such as communication and teamwork — is stunted. These are the very skills that respondents across all ASEAN countries agree are most critical for building a truly integrated workforce, and their absence means that **28%** of the potential benefits from genuine cross-border collaboration go unrealized.

   This is where AI can make a real difference. With **64% of the public** already turning to the internet and digital platforms for cultural information, the demand for a digital-first solution is clear. By leveraging Large Language Models (LLMs) to surface and prioritize

local wisdom — which **54% of respondents** agree significantly shapes the development of cultural products — Culbi builds the Intercultural Communicative Competence (ICC) that people need to connect meaningfully across borders, and to overcome the market access barriers currently faced by **68% of regional players**.

2. Developed Solution

To address the persistent cultural illiteracy and fragmented regional identity across ASEAN, we developed **Culbi**, a mobile application that transforms everyday exploration and travel into an immersive, AI-powered cultural learning experience. Rather than delivering cultural education through textbooks or formal curricula, Culbi embeds intercultural understanding into activities users already enjoy — discovering new places, planning trips, connecting with peers, and collecting achievements.

How Culbi Addresses the Problem

The core challenge is not a lack of information about ASEAN cultures — it is a lack of meaningful, personal exposure to them. Young people across the region hold superficial or stereotyped perceptions of their neighbours, and existing educational tools fail to translate cultural knowledge into lived experience. Culbi tackles this through three principles: making culture discoverable, making engagement rewarding, and making connection effortless.

Key Features:

2.1. Destinations

Culbi's Destinations feature presents cultural and tourist landmarks across ASEAN in an intuitive horizontal card interface. Tapping into any destination reveals a rich detail page: high-quality photos, star ratings and reviews sourced from the Google Maps API, a concise location description, and AI-generated cultural insights specific to that place. Crucially, each destination includes a **Cultural Ethics** section — outlining what is required, what is recommended, and a practical pro tip for respectful conduct at that location. Users can also directly ask **Culbi**, the in-app AI chatbot, any follow-up questions about the destination. This feature directly combats one of the most common intercultural pitfalls: arriving somewhere without understanding the unwritten rules.

2.2. Events

The Events feature aggregates cultural, community, and social events happening across ASEAN, complete with dates, venues, participant expectations, and geographic reach. Each event listing is enriched with an AI-generated fun fact, helping users see regional events not just as things to attend, but as windows into the living cultures of their neighbours.

2.3. Groups

Users can create location-based groups tied to specific coordinates on the map. Public groups are discoverable by all users through the Groups Explore section, displaying the group name, description, category, participant count, capacity, and membership. This feature encourages users from different regions and countries to form communities around shared cultural interests — fostering the peer-to-peer, cross-border connections that are foundational to genuine intercultural understanding.

2.4. Discovery Maps

Discovery Maps gives users an interactive map experience where ASEAN landmarks are displayed as rich markers containing images, names, locations, and reviews. Users can also drag and drop to create new groups directly on the map. Standard map settings such as satellite view toggling are included, making navigation intuitive for all users regardless of familiarity with the region.

2.5. Missions

Culbi's mission system drives consistent engagement through a competitive point-based structure. Users earn points by completing culturally meaningful tasks — such as sending messages to another users, visiting new landmarks, or exploring new regions. Points accumulate to build a user's leaderboard ranking, turning cultural exploration into a healthy, motivating competition.

2.6. Trips

The Trips feature combines an interactive map with an in-app camera tool. When users photograph a landmark, Culbi generates a **Share Photo Card** — a beautifully formatted shareable image — and a **Fun Fact Flash Card** revealing hidden cultural and historical insights about that location. Each photograph earns points redeemable for the card gacha system, directly rewarding real-world cultural exploration with in-app progression.

2.7. Trip Planner

The Trip Planner allows users to build and manage both upcoming and completed itineraries. Trips can be created manually or collaboratively with Culbi, the AI assistant, which tailors recommendations to the user's preferences, interests, and home region. Trip plans can also be shared and co-edited with friends or groups, enabling collaborative cultural exploration across borders.

2.8. Culbi Chatbot

At the heart of the application is **Culbi**, an intelligent conversational AI assistant that serves multiple roles: trip planner, language tutor, cultural interpreter, translator, and general companion. Culbi can browse the internet in real time, read documents and webpage content, open Google Maps, and personalize responses based on the user's username and home region. Whether a user needs to understand a local custom, translate a phrase, or simply have a conversation with someone who knows the region well, Culbi is always available. This directly addresses the ICC gap identified in the problem statement — giving users a real-time, contextually aware mediator for intercultural communication.

2.9. Messages

Culbi includes a full-featured messaging system supporting direct messages and group chats. This enables organic, ongoing relationships between users from different ASEAN nations — moving cultural exchange from a one-time event into a continuous dialogue.

2.10. Profile & Collections

Each user's Profile houses their personal **Collections**: a growing library of landmark cards and digital passports earned through their explorations. Landmark cards are unlocked through a gacha system using points earned in-app, and are categorized by rarity tiers — **Common, Rare, Epic, Legend, Mythic, and Secret** — determined by how remote and difficult to reach the associated landmark is. The further and more culturally distinct the location, the greater the chance of earning a high-rarity card. This design intentionally incentivizes users to explore beyond their comfort zones and discover the deeper, lesser-known corners of ASEAN culture.

2.11. Passport System

When users achieve milestones in a specific region — such as photographing 1 landmarks in Pontianak, Indonesia — they earn a **Digital Passport** for that territory. This passport system creates a tangible sense of cultural achievement and regional identity, making the ASEAN community feel personally earned rather than abstractly defined.

3. Technologies and Tools Used

Culbi is built on a carefully chosen stack where every technology decision serves the product's core mission: real-time, AI-powered intercultural discovery at scale across ASEAN.

3.1. The Mobile Client runs on React Native and Expo

The entire mobile experience is built in TypeScript on React Native 0.81.5, giving Culbi a single codebase that ships natively to both iOS and Android without compromise. Expo SDK 54 manages the build pipeline, over-the-air updates, and access to native device APIs, while Expo Router 6 provides file-based navigation so every screen maps directly to a file — keeping the project structure predictable as the feature set grows.

Styling is handled by NativeWind 4, which brings Tailwind CSS utility classes to React Native and ensures visual consistency across the entire app through a single design token system defined in tailwind.config.js and constants/style.ts. For animations that need to feel truly native, React Native Reanimated 3 runs transitions on the UI thread — keeping card reveals, scroll effects, and mission progress bars smooth even under load. The Share Photo Card and landmark card visuals are rendered with React Native Skia, which provides GPU-accelerated canvas drawing capable of the gradients and rarity shimmer effects that standard stylesheets cannot produce.

3.2. Maps and location are powered by Google

Discovery Maps is built on react-native-maps 1.20, wrapping the native Google Maps SDK for tile rendering, satellite imagery, and gesture handling. Nearby landmark search, place photos, ratings, and reviews are all sourced from the Google Places API, while expo-location feeds the device's GPS coordinates into both the landmarks_near geospatial query and the Culbi chatbot's location context. At lower zoom levels, react-native-map-clustering automatically groups dense landmark markers to keep the map readable across all of ASEAN.

3.3. The AI layer is built entirely on Google Gemini

All AI capabilities in Culbi run through Google Gemini 2.5 Flash, chosen for its low latency, native multimodal vision, and built-in tool-use support. The google-generativeai Python SDK connects the FastAPI backend to Gemini's full suite of capabilities: Google Search grounding for real-time factual answers, a URL Context tool for reading webpages and documents, Code Execution for sandboxed calculations, and File Search (RAG) for retrieval over Culbi's cultural knowledge base.

The backend itself is a Python 3.12 FastAPI service running on Uvicorn, with Pydantic enforcing typed validation on every request and response. Because Gemini SDK calls are blocking I/O, they are offloaded to a ThreadPoolExecutor with 8 workers — keeping the async event loop free to handle concurrent HTTP requests without stalling.

3.4. Supabase serves as the data and auth backbone

All persistent data lives in Supabase, which consolidates hosted PostgreSQL 15, authentication, real-time messaging, and object storage under one platform. The database uses PostGIS for indexed geography queries, enabling the

landmarks_near RPC to perform radius searches directly in the database rather than computing distances on the client. Business-critical operations — XP awards, mission completions, leaderboard rankings, group creation — are implemented as SECURITY DEFINER PostgreSQL RPC functions, ensuring they are atomic, server-side, and protected by Row-Level Security regardless of client behaviour.

Authentication is anonymous-first: every user receives a stable Supabase session on first launch, and if they later sign in via Google OAuth, Supabase links the Google identity to the existing session — preserving all their history without changing their user ID. For messaging, Supabase Realtime opens a WebSocket subscription directly from the mobile client, delivering new messages instantly without polling.

3.5. The events pipeline runs on PredictHQ

Cultural and community events across ASEAN are sourced from PredictHQ, which provides structured event data including dates, venues, attendance predictions, and geographic scope. A scheduled scraper pipeline — built in Python across predicthq_client.py, transformer.py, and db.py — fetches, normalises, and upserts this data into the events table in Supabase. The mobile app reads directly from that table at runtime and has no knowledge of PredictHQ, keeping the client simple and the data always fresh.

3.6. Supporting libraries complete the experience

Camera capture for landmark detection is handled by expo-camera, with expo-file-system converting photos to base64 for upload to the AI backend. Generated Share Photo Cards are saved to the device via expo-media-library and shared through expo-sharing. In-app gestures rely on react-native-gesture-handler and @gorhom/bottom-sheet for performant swipe interactions and snap-point overlays, while react-native-gifted-chat provides the messaging UI customised with NativeWind tokens. Tactile feedback on card reveals and mission completions is delivered through expo-haptics, reinforcing the reward loop with a physical response. The app's typography is built entirely on Inter, loaded once at the root layout via @expo-google-fonts/inter.

4. Solution Architecture & Workflow

Culbi is built around four major components that work together seamlessly: the Mobile Client, the AI Backend, the Data Platform, and a set of External APIs. Understanding how these four layers interact explains the entire system.

4.1. The Mobile Client serves as the front door for every user interaction. The app is built with React Native and Expo, running on both iOS and Android, with a file-based routing system that maps each screen directly to its own file. All screens share two global pieces of state — authentication (who the user is) and trip context (the current trip being planned) — injected at the root layout and

available everywhere in the app. Rather than screens talking directly to the database or backend, every data concern is encapsulated in a dedicated custom hook — such as `useLandmarkDetail`, `useEvents`, and `useGeminiChat` — keeping screens clean and data-fetching logic reusable across the app.

4.2. **Supabase** serves as the primary data store and authentication backbone. Almost all persistent data lives here — built on a hosted PostgreSQL database with geospatial support, Row-Level Security, real-time WebSocket messaging, and managed authentication. Authentication is anonymous-first: every user gets a stable session the moment the app opens, and if they later sign in with Google, their entire history — chats, trips, and XP — carries over seamlessly. For any operation requiring complex logic, such as geospatial landmark lookup, mission progress tracking, or leaderboard ranking, the app calls server-side stored procedures rather than raw SQL, keeping business logic protected and centralized. Real-time messaging is powered by a WebSocket subscription that delivers new messages instantly, without polling.

4.3. **The AI Backend** is the intelligence layer, built as a FastAPI service in Python. The mobile app never calls Google Gemini directly — all AI work is routed through this backend, which keeps API keys server-side and provides a single place to manage prompts, tool routing, and response shaping. At its core is a dispatcher that routes each request to the appropriate Gemini-native tool: web search, Google Maps, URL reading, document retrieval, or code execution. Three specialized endpoints extend this core — `/landmark-detection` identifies real-world landmarks from camera photos using Gemini's multimodal vision and returns cultural notes and confidence scores; `/trip-concierge` receives a full itinerary and a natural-language instruction, then proposes structured changes that users can accept or discard before anything is saved; and `/chat` powers the Culbi conversational assistant, which receives the user's full conversation history, profile, and location to deliver contextually aware, tool-augmented responses.

4.4. **External APIs** extend the platform's data surface at key points. Google Maps and Places API is called directly from the mobile client for low-latency map rendering, nearby place search, and reviews. Google OAuth handles sign-in securely on-device, linking identities to existing sessions via Supabase. PredictHQ powers the Events feature entirely through an offline pipeline — a scheduled job fetches upcoming ASEAN events, normalizes them, and upserts them into the database, so the mobile app simply reads from a clean, ready table at runtime.

In a typical session, these four layers work in concert. When a user opens Culbi, the app silently establishes a session and simultaneously fetches nearby landmarks, upcoming events, and public groups before the first screen renders. Tapping a landmark pulls its cultural rules, photos, and reviews from the database, then requests AI-generated insights from the backend on the fly. Opening the camera and photographing a landmark triggers the full gamification loop — landmark identification, rarity roll, card collection, XP update, and mission progress — all in a single coordinated sequence from one photo capture. This layered design ensures that each component has a single, clear responsibility, and can be scaled or improved independently as Culbi grows.

5. Feasibility & Challenges

5.1. Feasibility

Culbi is ambitious in scope, but every major technical decision was made deliberately to reduce implementation risk rather than compound it. The result is a system that can be built incrementally, validated layer by layer, and scaled without architectural rewrites.

The foundation starts with managed services. Supabase alone replaces what would otherwise be four separate engineering concerns — authentication, a relational database, file storage, and real-time messaging — all wired together through a single typed client library. Expo similarly eliminates the need to maintain separate iOS and Android native codebases. This compression of infrastructure work means the team's engineering effort goes into product features, not plumbing.

The layered architecture reinforces this. Because the mobile client, AI backend, and database communicate through well-defined interfaces, each layer can be built and validated independently. The Destinations feature can be fully working before the chatbot is written. The AI backend can be tested with a single curl command without a running app. The events scraper runs entirely on its own schedule, invisible to the client. Incremental delivery is not just possible — it is the natural shape of the system.

Business-critical logic is centralised in PostgreSQL RPC functions rather than distributed across client code. XP awards, mission completions, and geospatial queries all live as SECURITY DEFINER functions with Row-Level Security enforcement. They are testable with plain SQL, immune to client-side tampering, and ready to serve a second client — a web app or admin panel — without reimplementation.

On the AI side, Google Gemini 2.5 Flash was chosen precisely because it treats multimodal vision, grounded web search, URL reading, and code execution as first-class native tools rather than bolt-on integrations. The landmark detection pipeline, chatbot, and trip concierge all run on the same model through the same SDK — a meaningful reduction in operational complexity compared to assembling multiple specialised ML services.

Finally, the anonymous-first authentication model eliminates the drop-off point that kills most consumer app funnels. Users see the full app immediately, with no sign-up wall.

Their session is stable from first launch, and linking a Google identity later carries all history forward seamlessly — a pattern proven at scale by apps like TikTok and Google Maps.

The table below summarises the estimated implementation difficulty per feature area, assessed against the actual technical complexity involved rather than feature surface area.

Feature Area	**Difficulty**	**Rationale**
Auth + onboarding	Low	Supabase anonymous auth + Google Sign-In are well-documented; identity linking is a single API call
Destinations + landmark content	Low	Standard CRUD with Supabase; PostGIS RPC is pre-written
Discovery Maps	Medium	Marker clustering and viewport-bound group queries require performance tuning on large datasets
Culbi Chatbot	Medium	Gemini multi-tool chat is straightforward; complexity lies in prompt engineering and context management
Landmark Detection + Cards	Medium-High	Multi-step pipeline — base64 upload, Gemini vision, Skia rendering, rarity roll — with several failure points
Trip Planner + AI Concierge	Medium-High	PROPOSE_CHANGES schema enforcement and ghost-stop UI state management are non-trivial
Missions + XP System	Medium	Atomic RPC design is complete; the challenge is wiring increment_mission_progress consistently across all triggers
Messaging (DM + Group)	Medium	Supabase Realtime handles delivery; complexity is UI correctness for read receipts and image upload lifecycle
Passport + Gacha System	Medium	Rarity engine is compact; complexity is Skia card design and milestone tracking trigger logic
Events Pipeline	Low	Self-contained Python scraper; PredictHQ API is well-documented with a Python SDK

5.2. Challenges

Building Culbi involves seven clearly defined engineering and operational challenges. Each has been anticipated and mitigated by design decisions already present in the architecture.

Gemini response latency on mobile is the most user-visible risk. Calls involving multiple tools — search, URL context, and vision together — can take three to eight seconds on a mobile connection. Culbi addresses this at three levels: the FastAPI backend processes up to eight concurrent requests through a ThreadPoolExecutor so no user blocks another; the mobile client shows immediate skeleton and shimmer loading states so the UI never appears frozen; and landmark enrichment is split into two independent calls, surfacing the detection result first while the three Cultural Flashcards load in the background. For the chatbot specifically, Gemini's streaming API can begin rendering token-by-token output before the full response is complete.

Regional ASEAN landmark coverage gaps present a data quality challenge. Many culturally significant landmarks in rural Indonesia, interior Borneo, or rural Myanmar are underrepresented in Google Maps and standard ML training data. The detection prompt explicitly instructs Gemini to use Google Search grounding as its primary verification source rather than parametric knowledge, catching many regional landmarks the model alone would miss. Every detection result also includes a confidence_score field, allowing the app to surface a low-confidence warning and prompt the user to confirm. The landmarks table accepts sovereign_custom photos and descriptions, and the seed_landmarks.py and backfill_photos.py backend tools allow the team to proactively enrich known coverage gaps before users encounter them.

Real-time messaging scalability is a concern as group activity grows. Supabase Realtime uses PostgreSQL logical replication to broadcast row changes over WebSockets, which is highly effective at low-to-medium traffic but could stress connection limits as the number of active groups scales. Each client subscribes only to the specific group_id or conversation_id currently open — not a global stream — limiting fan-out at the source. Message pagination ensures only recent messages are held in memory, with historical messages fetched on scroll. If Supabase Realtime does become a bottleneck at scale, the messaging tables are cleanly isolated and can be migrated to a dedicated real-time service with only hook-level changes on the client.

XP and mission integrity could be compromised by duplicate calls from network retries or optimistic UI updates. The increment_mission_progress RPC is an atomic PostgreSQL transaction — the counter increment and completion check happen together, so concurrent calls cannot both trigger the completion branch. The user_xp_ledger is append-only and immutable by design, and claim_mission_reward is idempotent — calling it twice after the first claim has no effect. RLS policies block all direct client mutations to these tables, meaning every XP change must pass through the RPCs.

App store compliance for camera features, AI-generated content, and the gacha system requires careful handling before submission. Camera access follows runtime permission guidelines with a clear in-app explanation of purpose. The gacha system awards cards using engagement-earned in-app points rather than real-money purchases, which avoids the loot box regulatory category in most jurisdictions — though this should be verified by legal review before launch. AI-generated content throughout the app — cultural facts, trip suggestions, chatbot responses — should carry visible "AI-generated" labels in the UI, in line with Apple and Google's emerging content policies.

PredictHQ data freshness and coverage could leave the Events feature incomplete for smaller ASEAN cities and community-level cultural events. The scraper uses phq_id as its upsert key, making re-runs safe and idempotent at any time. The ScrapeResult dataclass tracks fetched, upserted, skipped, and error counts per country, enabling monitoring and alerting on job health. For local events outside PredictHQ's coverage, a future sovereign_custom event source — following the same pattern already established for landmark photos — would allow the team or trusted users to submit events directly into the events table.

Cold start and onboarding drop-off is the final risk. The app's core value — nearby landmarks, regional groups, relevant events — is geographically dependent, and a user who skips onboarding or denies location permissions will see a sparse home screen. The anonymous-first session allows exploration before any profile setup, and a DEFAULT_REGION constant centred on Kuching, Sarawak ensures the map and landmark cards are never empty on first open. Onboarding screens are non-blocking overlays rather than hard gates, and location permission is requested contextually — when opening the map or tapping "nearby landmarks" — rather than on app launch, which significantly improves grant rates on both iOS and Android.

6. Expected Outcome & Impact

6.1. The Problem Culbi Solves

Before Culbi, cultural exploration in ASEAN was fragmented across tools that were never designed for the purpose.

Generic travel apps like Google Maps and TripAdvisor surface place names, ratings, and opening hours — but they do not explain why a temple has a dress code, what a local name means phonetically, or what a lesser-known community festival represents in its regional context. A traveler arrives informed about logistics but culturally unprepared. General-purpose AI assistants fare no better: asking ChatGPT or Gemini about ASEAN landmarks yields accurate but surface-level results, with no awareness of the user's current location, travel itinerary, or what they have already visited. They cannot look at a photo and

identify the Kek Lok Si Pagoda while simultaneously explaining Buddhist temple conduct, or propose a reordered itinerary when the user asks to swap two days.

Trip planning, meanwhile, remains scattered across Pinterest boards, Google Docs, Maps bookmarks, and WhatsApp threads — with no single tool that holds the full plan, supports AI-assisted editing, accommodates collaborators, and persists the history of what was visited in a meaningful way. This problem is compounded by a deeper visibility gap: mainstream apps are trained on high-traffic tourist circuits in Bali, Bangkok, Kuala Lumpur, and Singapore. Indigenous sacred sites in interior Borneo, community shrines in rural Vietnam, traditional weaving villages in Sulawesi, and pre-colonial structures in the southern Philippines are absent from or poorly represented in standard ML models and curated databases. And even for travelers who do reach these places, there is no engagement loop to record, reflect on, or share the journey in a culturally meaningful way — producing checkbox tourism that is physically present but culturally absent.
Culbi is designed to address each of these gaps simultaneously.

6.2. Expected Benefits

- **For the traveler experience**

  Rather than reading a dozen blog posts before visiting a heritage site, users open Culbi and find structured cultural rules — categorised as Mandatory, Recommended, or Pro-Tip — alongside AI-generated insights grounded in real-time Google Search, and conversational answers from a chatbot that knows exactly where they are and what they have already visited. Hours of pre-trip research are compressed into in-the-moment contextual guidance delivered at precisely the right time.

  For landmarks that standard apps cannot identify, Culbi's camera-powered detection fills the gap. A traveler can point their phone at an unfamiliar stone carving, roadside shrine, or heritage building and receive an immediate identification — including its local name, historical context, and cultural significance — even if it does not appear on Google Maps. The Gemini multimodal vision pipeline, grounded with live Google Search, covers ASEAN landmarks that no standard image recognition API handles correctly.

  Trip planning is consolidated into a single conversational interface. A user types "I have 5 days in Penang, focus on heritage and street art" and receives a structured itinerary linked to Culbi's cultural database. When plans change, they message the AI Concierge in natural language and see proposed edits rendered as ghost stops — accepted or rejected in one tap — while collaborators follow along in real time.

Underlying all of this is a gamification system that rewards depth rather than breadth. The XP ledger, daily missions, passport stamps, and rarity card collection create a loop that mechanically incentivises seeking out less-visited, more culturally significant sites. A user who photographs a legendary-rarity landmark earns significantly more than one who visits the obvious honeypots — turning cultural curiosity into a sustained, progressive habit.

- **For local communities and cultural heritage**
  Culbi creates a direct path to digital representation for heritage sites that have never appeared in any app. The sovereign_custom photo and review source, combined with the AI recognition pipeline, means a community can work with the Culbi team to ensure their sacred site is correctly identified, named in the local language, and accompanied by culturally appropriate guidance — rather than being misidentified or absent entirely.
  The three-tier landmark_rules system gives cultural custodians a structured channel to communicate visitor expectations. Rather than relying on a sign at the gate that tourists may not read, mandatory and recommended rules appear in the app before the user even arrives — normalising respectful behaviour as part of the discovery experience, not an afterthought.
  Community events gain reach through the PredictHQ pipeline, which surfaces local festivals, ceremonies, and cultural markets to an audience of culturally engaged travelers who would not have found them otherwise. And when the app surfaces heritage sites in smaller towns and rural regions alongside the major tourist circuits, it nudges traveler foot traffic — and spending — toward communities that currently receive little benefit from regional tourism. This is the mechanism by which a digital product creates tangible economic impact beyond the screen.

- **For the broader cultural tourism ecosystem**
  Culbi's database architecture — landmarks, rules, cultural facts, and photos organised by source — creates a structured foundation for tourism boards, UNESCO-affiliated heritage bodies, and cultural NGOs to contribute verified, authoritative content that reaches travelers at the exact moment of discovery. This is significantly more impactful than a brochure or a static website.
  The app's engagement data — which landmarks are photographed, which cultural rules are viewed, which events are attended — provides aggregate, privacy-safe signal about how travelers engage with cultural sites. This data is currently unavailable to cultural institutions, which rely on ticket sales and visitor counts that miss the full picture of engagement.

Finally, by demonstrating that Gemini vision can correctly identify ASEAN landmarks with a confidence score, by labelling AI-generated content transparently, and by grounding cultural facts in real-time Search rather than LLM hallucination, Culbi sets a responsible AI deployment pattern that other regional tourism products can learn from and build on.

7. **Who Benefits**

Culturally curious travelers gain AI-guided discovery of landmarks they would never find independently, with cultural context delivered at the right moment and trip planning consolidated in one tool. First-time ASEAN visitors receive etiquette guidance that prevents disrespectful behaviour before it happens, backed by a knowledgeable, non-judgmental chatbot available 24/7. Repeat visitors find a reason to seek out new and lesser-known sites through the gamification system — XP, passports, and rare cards create ongoing motivation that outlasts the first trip.

Backpackers and independent travelers get a full-featured AI trip planner that replaces paid tours and expensive travel agents for itinerary construction. Travel content creators benefit from the Skia-rendered Share Photo Card — a visually distinct, rarity-labelled artefact that stands apart from generic screenshots. Local communities hosting heritage sites gain digital representation for the first time, with culturally appropriate visitor guidelines surfaced proactively to arriving travelers.

Cultural NGOs and heritage bodies gain a structured platform to contribute authoritative content and reach engaged audiences. ASEAN tourism boards gain a distribution channel for lesser-known destinations and community events to high-intent travelers. And collaborative travelers — friends, families, and groups — gain a shared trip planning space with real-time editing and group chat tied to physical location.

Summary of Expected Outcomes

Culbi's expected impact spans five interconnected areas. In landmark discoverability, heritage sites outside major tourist circuits become findable via AI visual recognition — transforming a smartphone camera into a cultural guide for the first time. In cultural understanding, mandatory and recommended rules surfaced before visits create a measurable shift toward respectful behaviour at sacred sites. In geographic spread, users guided to lesser-known destinations disperse economic benefit to smaller communities beyond the established tourist corridors of Bali, Bangkok, Kuala Lumpur, and Singapore.

In trip planning efficiency, the AI Concierge replaces three to four fragmented tools — Docs, Maps, Pinterest, WhatsApp — with a single conversational interface that holds the full plan from inspiration to completion. And in user retention, the interlocking

gamification system — XP, missions, daily challenges, passport stamps, and card rarity — converts one-time visitors into long-term cultural explorers with a reason to return.
Taken together, these outcomes position Culbi not just as a travel tool, but as infrastructure for a more culturally literate, economically distributed, and responsibly documented ASEAN tourism ecosystem.

8. Marketing Potential

Culbi operates at the intersection of cultural tourism, AI-powered education, and social discovery — three sectors that are growing simultaneously across ASEAN and that share a common, underserved audience: digitally-native young people who are curious about the region but lack a structured, engaging way to explore it. The marketing opportunity is therefore not just about acquiring users; it is about positioning Culbi as the defining cultural platform for the next generation of ASEAN citizens.

The addressable audience is large and already digitally primed. ASEAN's youth population — broadly defined as those aged 15 to 35 — represents hundreds of millions of individuals across ten nations. As established in the problem statement, 64% of this demographic already turns to the internet and digital platforms as their primary source of cultural information. This is a population that does not need to be taught to use a digital tool for cultural discovery; it simply needs a tool worth using. Culbi is designed to be exactly that.

Tourism partnerships create a high-leverage distribution channel. Travel agencies across ASEAN — particularly those specialising in cultural, heritage, and community-based tourism — represent a natural first partnership tier. A traveller who books an itinerary through an agency could arrive with Culbi pre-installed, pre-loaded with their destination's landmarks, cultural rules, and the trip plan tailored to their package. This model benefits the agency — which gains an AI-powered value-add for their clients at no additional operational cost — and Culbi, which acquires an engaged user at the highest-intent moment: the days immediately before and during travel. Formalised partnerships with regional agency networks in Malaysia, Indonesia, Thailand, and the Philippines would establish Culbi's presence within the established tourism supply chain rather than competing against it.

Destination partnerships unlock institutional credibility and co-marketing reach. Heritage sites, cultural parks, national museums, and UNESCO-listed landmarks across ASEAN face a consistent challenge: increasing visitor engagement beyond a surface-level visit. Culbi's landmark rules system, AI-generated cultural insights, camera detection, and gacha card rewards are precisely the mechanisms that transform passive visitors into active participants. A partnership with a cultural destination — for instance, Angkor Wat Heritage Trust, the Borobudur Authority, or George Town's heritage

conservation board — could take the form of a co-branded landmark card, a site-exclusive mission, or a Digital Passport milestone for that location. These partnerships serve dual marketing functions: they give Culbi an institutional stamp of authenticity that accelerates user trust, and they give the destination a compelling story for their own communications about how they are using technology to deepen visitor engagement.

National tourism boards represent the highest-impact institutional partner tier. Tourism Malaysia, TAT Thailand, Wonderful Indonesia, Visit Philippines, and their counterparts are each running active campaigns to attract younger, culturally motivated travellers. Culbi's data layer — aggregated, privacy-safe signals about which landmarks are visited, which cultural events are attended, and which regions users explore — is precisely the kind of engagement intelligence that tourism boards currently lack. A memorandum of understanding with even two or three national tourism boards would simultaneously unlock government-backed promotional reach, grant access to authoritative content for the sovereign_custom landmark source, and provide the institutional relationships needed to navigate app store approvals and regulatory requirements at scale across ASEAN.

The app's social mechanics generate organic, zero-cost acquisition. The Share Photo Card — a Skia-rendered, rarity-labelled artefact produced every time a user photographs a landmark — is a native viral loop. When a user shares their Mythic-rarity card for a remote longhouse in interior Sarawak to Instagram or WhatsApp, the accompanying deep link either opens the exact landmark screen in Culbi for existing users, or routes new users to the app store. This turns every cultural discovery into a referral moment, and more importantly, it does so in a context where the referrer is enthusiastically sharing something they are genuinely proud of — a far higher-trust signal than a paid advertisement. The same principle applies to Digital Passport: a user who earns the Sarawak Heritage Passport has social motivation to display it, and every display is a product impression that carries the implicit endorsement of a real traveller.

The competitive positioning is defensible. Generic travel apps do not build cultural literacy; they surface logistics. General-purpose AI assistants do not know where a user is standing, what they have already visited, or what a specific landmark's dress code requires. No existing product combines camera-powered landmark detection, AI-generated cultural ethics guidance, a cross-border social layer, and a progressive gamification system in a single mobile experience purpose-built for ASEAN. This specificity — building deeply for one region rather than shallowly for the world — is both Culbi's brand and its moat. A user who has accumulated fifty landmark cards, earned three regional passports, and built cross-border friendships through Culbi's group system has switching costs that no generic travel app can replicate.

The long-term commercial model aligns incentives across all stakeholders. Culbi's v2.0 roadmap introduces affiliate booking integration — surface a Klook or GetYourGuide link when the AI Concierge adds a guided temple tour to a trip — and a tourism board content API that allows institutions to push verified content into the platform in exchange for featured placement. These revenue streams are structurally aligned with user value: the user gets a better itinerary, the booking partner acquires a high-intent customer, the cultural institution reaches an engaged audience, and Culbi earns a margin on each transaction without introducing advertising or data monetisation that would erode the trust the platform depends on. Culbi's marketing potential is therefore not a campaign to be executed — it is a compound flywheel: the more landmarks users discover, the more cards they share; the more partnerships Culbi signs, the more authoritative its content becomes; and the more culturally credible the platform grows, the more institutions want to be part of it.

9. Future Enhancements

The current architecture is deliberately scoped to a feature-complete v1 that can be delivered incrementally and validated with real users. The enhancements below are ordered by strategic priority — earlier items build directly on the existing foundation; later items represent larger platform expansions.

9.1. AI & Intelligence Layer

The most immediate AI enhancement is streaming responses from the /landmark-detection endpoint. Currently the endpoint returns a fully-formed payload only after the complete Gemini call finishes. Adding Server-Sent Events would allow the mobile client to display the landmark name the moment it is generated, with confidence score, cultural facts, and flashcards filling in progressively — reducing perceived latency from several seconds to near-instant initial feedback.

Beyond speed, the Trip Concierge can be made significantly smarter through personalised itinerary scoring. Rather than waiting for a user's plain-language instruction, the concierge would automatically score existing itineraries against the user's travel profile — preferred pace, landmark categories they respond to historically, and regional expertise level — and proactively suggest optimisations without being asked.

Two further enhancements extend Culbi's input modalities. Voice integration, built on expo-speech or the Gemini audio API, would let users dictate notes and commands hands-free while exploring — a voice note attached to a trip stop could be transcribed, semantically tagged, and surfaced as a future recommendation when planning similar trips. For rural ASEAN travel where connectivity is unreliable, a lightweight on-device model such as Google's Gemma via MediaPipe would handle basic landmark identification locally, with high-confidence or complex detections continuing to route through the FastAPI backend.

Finally, contextual push notifications would allow the backend — via Supabase Edge Functions and push tokens — to proactively surface time-sensitive prompts when a user is near an unvisited landmark or approaching a day where a cultural event is happening along their planned route.

9.2. Gamification & Engagement

The rarity card collection, currently private to each user, becomes a social feature in the next evolution: users could showcase their rarest cards on a public profile wall and trade duplicate cards with other collectors, turning the collection into a social currency that rewards deep exploration and drives inter-user connection.

Seasonal challenge events would align Culbi's gamification with the actual cultural calendar — a Hari Raya Heritage Trail during Ramadan, a Water Festival Circuit during Songkran, a Batik Artisan Hunt tied to UNESCO Batik Day. These limited-time challenges create shared community goals and drive concentrated engagement spikes anchored to real cultural moments rather than abstract daily resets.

Groups would evolve from conversation spaces into mission-running squads. A group could set a shared objective — photographing ten colonial-era buildings in Malacca over a weekend — track collective progress on a shared mission board. Above the existing explorer ranks, a prestige layer would reward highly active users with cosmetic profile frames, exclusive card art variants, and curator roles that allow them to propose new landmark content for review.

9.3. Content & Data

The most impactful content enhancement is opening a structured community submission flow — allowing users to propose new landmarks, correct existing local names, and submit photos directly from the app. Submissions would enter a moderation queue before being merged, dramatically accelerating coverage of rural heritage sites without requiring the core team to seed everything manually.

For authoritative content at scale, an authenticated content API would allow national tourism boards — Tourism Malaysia, TAT Thailand, Wonderful Indonesia, and others — to push verified landmark records, official event listings, and regulatory notices directly into the Culbi database. Official content would be visually differentiated with a verification badge, establishing Culbi as a trusted distribution channel for institutional cultural information.

Two further content enhancements deepen the on-site experience. AI-generated audio walking tours — short narrated guides of one to three minutes, generated once via Gemini and Google Cloud TTS, then stored and streamed from Supabase Storage — would give users a guided experience at heritage sites without requiring them to look at their screen. A landmark condition and accessibility reporting feature would let users flag

temporary closures, construction, or access restrictions, with aggregated reports updating a status field on the landmarks table to give future visitors reliable real-time information.

9.4. Social & Community

The Share Photo Card already produces a high-quality shareable image. Embedding a deep link into that card — one that opens the correct landmark detail screen in Culbi, or triggers an app store install prompt if Culbi is not yet installed — would create a viral acquisition loop driven entirely by user-generated cultural content shared across Instagram, WhatsApp, and other platforms.

A verified local expert programme would allow vetted guides, archaeologists, and cultural practitioners to create expert profiles within the app. Their contributions to landmark facts, cultural rules, and audio guides — building trust for culturally sensitive content and laying a marketplace-ready foundation for future guided experience bookings.

At the landmark level, a structured community Q&A layer would let users post and answer practical questions — "Is the inner shrine accessible to non-Hindus?" — with the most helpful responses voted to the top. This produces a living, community-maintained knowledge layer that complements AI-generated content with practical first-hand experience.

9.5. Trip Planning & Logistics

The Trip Planner currently holds ordered stops but provides no routing or transport guidance. Integrating the Google Directions API — and Grab or Gojek APIs where applicable — would add estimated travel times between stops, recommended transport modes, and fare estimates, turning a destination list into a fully executable itinerary.

A thin monetisation layer would come through booking integration: when the AI Concierge adds a museum or guided temple tour, it could surface a direct booking shortcut via GetYourGuide, Klook, or Agoda affiliate deep links — aligning commercial intent with the user's existing planning workflow rather than interrupting it. Completed or planned trips could also be exported as a PDF itinerary, shareable link, or Google Calendar import, making the trip planner useful even for group members who do not have the app installed.

After a trip is marked complete, a personal analytics view — total distance covered, landmarks photographed, rarity distribution, most culturally diverse day, and a heatmap of visited locations — would transform the trip record from a static log into a reflective and shareable story.

9.6. Platform & Infrastructure

A lightweight React and Next.js web companion app — focused on trip planning and landmark discovery without camera or GPS features — would expand Culbi's

addressable audience to desktop planners who research and book trips months in advance. The existing Supabase schema, RPC layer, and FastAPI backend are already web-compatible; only the React Native-specific UI components would need porting.

An internal admin content management portal would expose the sovereign_custom source field through a proper editorial UI, allowing the Culbi team to create, edit, and review landmark records, cultural rules, curated events, and community-submitted content without touching raw SQL. Time-sensitive backend operations — XP recalculation, mission resets at UTC midnight, leaderboard cache refreshes — would migrate from client-triggered RPCs to Supabase Edge Functions on database webhooks or cron schedules, removing the dependency on an active user session for background maintenance.

As Culbi expands across ASEAN, deploying the FastAPI backend across at minimum two regions — Singapore paired with Jakarta or Kuala Lumpur — would significantly reduce AI response latency for users in Indonesia, the Philippines, and Vietnam. Alongside this, instrumenting the backend with OpenTelemetry to track per-endpoint latency, Gemini tool usage, confidence score histograms, and error rates by country would allow the team to identify coverage gaps and latency outliers before users report them.

9.7. Enhancement Priority

The enhancements above are sequenced across four release phases. For the immediate post-launch period, the highest-value, lowest-complexity improvements are streaming landmark detection, community landmark submissions with moderation, and cross-platform sharing with deep links — all buildable directly on the existing architecture without significant infrastructure changes.

Version 1.2 introduces seasonal challenge events, transport and routing integration, and trip export to PDF and Google Calendar — features that deepen engagement and utility for users already comfortable with the core product. Version 1.3 brings audio walking tours and social card trading, which require new content pipelines and social mechanics but no fundamental architectural changes.

The v2.0 roadmap covers the larger platform expansions: the tourism board content API, the Next.js web companion app, the admin CMS portal, on-device offline AI via Gemma, multi-region backend deployment, and booking integration with commercial partners. These represent Culbi's evolution from a consumer app into a platform — one that institutions, communities, and partners can build on and contribute to.